

Questions with Answer Keys

MathonGo

Q1 - 2024 (01 Feb Shift 1)

A regular polygon of 6 sides is formed by bending a wire of length 4π meter. If an electric current of $4\pi\sqrt{3}$ A is flowing through the sides of the polygon, the magnetic field at the centre of the polygon would be $x \times 10^{-7}$ T. The value of x is _____

Q2 - 2024 (01 Feb Shift 2)

A moving coil galvanometer has 100 turns and each turn has an area of 2.0 cm^2 . The magnetic field produced by the magnet is 0.01 T and the deflection in the coil is 0.05 radian when a current of 10 mA is passed through it. The torsional constant of the suspension wire is $x \times 10^{-5} \text{ N - m/rad}$. The value of x is _____

Q3 - 2024 (27 Jan Shift 1)

Position of an ant (S in metres) moving in Y – Z plane is given by $S = 2t^2\hat{j} + 5\hat{k}$ (where t is in second). The magnitude and direction of velocity of the ant at $t = 1$ s will be :

- (1) 16 m/s in y-direction
- (2) 4 m/s in x-direction
- (3) 9 m/s in z-direction
- (4) 4 m/s in y-direction

Q4 - 2024 (27 Jan Shift 1)

A proton moving with a constant velocity passes through a region of space without any change in its velocity.

If $\vec{E}$ and $\vec{B}$ represent the electric and magnetic fields respectively, then the region of space may have :

- (A) $E = 0, B = 0$
- (B) $E = 0, B \neq 0$
- (C) $E \neq 0, B = 0$
- (D) $E \neq 0, B \neq 0$

Choose the most appropriate answer from the options given below :

- (1) (A), (B) and (C) only

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Questions with Answer Keys

MathonGo

(2) (A), (C) and (D) only

(3) (A), (B) and (D) only

(4) (B), (C) and (D) only

Q5 - 2024 (27 Jan Shift 1)

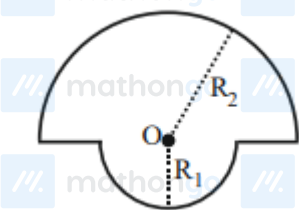
Two long, straight wires carry equal currents in opposite directions as shown in figure. The separation between the wires is 5.0 cm. The magnitude of the magnetic field at a point P midway between the wires is _____ μT

(Given : $\mu_0 = 4\pi \times 10^{-7} \text{TmA}^{-1}$)



Q6 - 2024 (27 Jan Shift 2)

The magnetic field at the centre of a wire loop formed by two semicircular wires of radii $R_1 = 2\pi\text{m}$ and $R_2 = 4\pi\text{m}$ carrying current $I = 4\text{ A}$ as per figure given below is $\alpha \times 10^{-7}\text{ T}$. The value of α is _____ (Centre O is common for all segments)



Q7 - 2024 (29 Jan Shift 1)

The magnetic potential due to a magnetic dipole at a point on its axis situated at a distance of 20 cm from its center is $1.5 \times 10^{-5} \text{Tm}$. The magnetic moment of the dipole is _____ Am^2 . (Given : $\frac{\mu_0}{4\pi} = 10^{-7} \text{TmA}^{-1}$)

Q8 - 2024 (29 Jan Shift 2)

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Questions with Answer Keys

MathonGo

Two particles X and Y having equal charges are being accelerated through the same potential difference.

Thereafter they enter normally in a region of uniform magnetic field and describes circular paths of radii R_1 and R_2 respectively. The mass ratio of X and Y is :

(1) $\left(\frac{R_2}{R_1}\right)^2$

(2) $\left(\frac{R_1}{R_2}\right)^2$

(3) $\left(\frac{R_1}{R_2}\right)$

(4) $\left(\frac{R_2}{R_1}\right)$

Q9 - 2024 (29 Jan Shift 2)

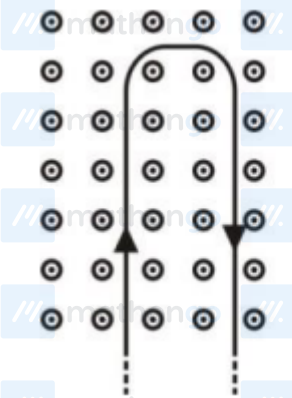
A charge of $4.0\mu\text{C}$ is moving with a velocity of $4.0 \times 10^6 \text{ ms}^{-1}$ along the positive y-axis under a magnetic field $\vec{B}$ of strength $(2\hat{k})$ T. The force acting on the charge is $x\hat{i}$ N. The value of x is _____

Q10 - 2024 (30 Jan Shift 2)

The current of 5 A flows in a square loop of sides 1 m is placed in air. The magnetic field at the centre of the loop is $X\sqrt{2} \times 10^{-7}$ T. The value of X is _____

Q11 - 2024 (31 Jan Shift 1)

A rigid wire consists of a semicircular portion of radius R and two straight sections. The wire is partially immersed in a perpendicular magnetic field $B = B_0\hat{j}$ as shown in figure. The magnetic force on the wire if it has a current i is :



Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Questions with Answer Keys

MathonGo

(1) $-iBR\hat{j}$

(2) $2iBR\hat{j}$

(3) $iBR\hat{j}$

(4) $-2iBR\hat{j}$

Q12 - 2024 (31 Jan Shift 1)

A coil is placed perpendicular to a magnetic field of 5000 T. When the field is changed to 3000 T in 2 s, an induced emf of 22 V is produced in the coil. If the diameter of the coil is 0.02 m, then the number of turns in the coil is :

(1) 7

(2) 70

(3) 35

(4) 140

Q13 - 2024 (31 Jan Shift 1)

An electron moves through a uniform magnetic field $\vec{B} = B_0\hat{i} + 2B_0\hat{j}T$. At a particular instant of time, the velocity of electron is $\vec{u} = 3\hat{i} + 5\hat{j}$ m/s. If the magnetic force acting on electron is $\vec{F} = 5ekN$, where e is the charge of electron, then the value of B_0 is _____ T.

Q14 - 2024 (31 Jan Shift 2)

A uniform magnetic field of 2×10^{-3} T acts along positive Y-direction. A rectangular loop of sides 20 cm and 10 cm with current of 5 A is Y – Z plane. The current is in anticlockwise sense with reference to negative X axis. Magnitude and direction of the torque is :

(1) 2×10^{-4} N – m along positive Z-direction

(2) 2×10^{-4} N – m along negative Z-direction

(3) 2×10^{-4} N – m along positive X-direction

(4) 2×10^{-4} N – m along positive Y-direction

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Questions with Answer Keys

MathonGo

Q15 - 2024 (31 Jan Shift 2)

Two circular coils P and Q of 100 turns each have same radius of π cm. The currents in P and R are 1 A and 2 A respectively. P and Q are placed with their planes mutually perpendicular with their centers coincide. The resultant magnetic field induction at the center of the coils is $\sqrt{x}$ mT, where $x =$ _____

[Use $\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$]

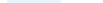
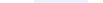
Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

MathonGo

Answer Key

Q1 (72) mathongo **Q2 (4)** **Q3 (4)** **Q4 (3)**

Q1 (72) mathongo  **Q2 (4)**  **Q3 (4)**  **Q4 (3)** 

Q5 (160) **Q6** (3) **Q7** (6) **Q8** (2)

Q5 (160) **Q6** (3) **Q7** (6) **Q8** (2)

Q9 (32) mathongo // **Q10 (40)** // mathongo **Q11 (4)** mathongo // **Q12 (2)** // mathongo

Q9 (32) mathongo // **Q10 (40)** // mathongo **Q11 (4)** mathongo // **Q12 (2)** // mathongo

Q13 (5) mathongo // Q14 (2) mathongo // Q15 (20) mathongo // mathongo

Q13 (5) mathongo **Q14 (2)** mathongo **Q15 (20)** mathongo

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

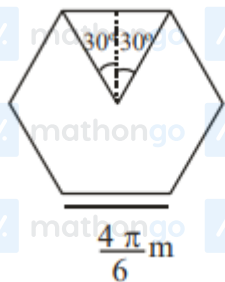
Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

Click here to download MARKS App

Solutions

MathonGo

Q1



$$B = 6 \left(\frac{\mu_0 I}{4\pi r} \right) (\sin 30^\circ + \sin 30^\circ)$$

$$= 6 \frac{10^{-7} \times 4\pi \sqrt{3}}{\left(\frac{\sqrt{3} \times 4\pi}{2 \times 6} \right)}$$

$$= 72 \times 10^{-7} \text{ T}$$

Q2

$$\tau = BINA \sin$$

$$C\theta = BINA \sin 90^\circ$$

$$C = \frac{BINA}{\theta} = \frac{0.01 \times 10 \times 10^{-3} \times 100 \times 2 \times 10^{-4}}{0.05}$$

$$= 4 \times 10^{-5} \text{ N-m/rad.}$$

$$x = 4$$

Q3

$$\vec{v} = \frac{d\vec{s}}{dt} = 4t\hat{j}$$

$$\text{At } t = 1 \text{ sec } \vec{v} = 4\hat{j}$$

Q4

Net force on particle must be zero i.e. $q\vec{E} + q\vec{V} \times \vec{B} = 0$

Possible cases are

$$(i) \vec{E} \& \vec{B} = 0$$

$$(ii) \vec{V} \times \vec{B} = 0, \vec{E} = 0$$

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Solutions

MathonGo

$$(iii) q\vec{E} = -q\vec{V} \times \vec{B}$$

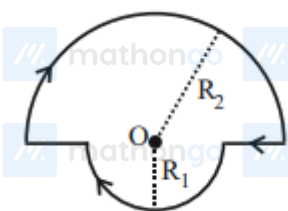
$$\vec{E} \neq 0 \& \vec{B} \neq 0$$

Q5

$$B = \left(\frac{\mu_0 i}{2\pi a} \right) \times 2 = \frac{4\pi \times 10^{-7} \times 10}{\pi \times \left(\frac{5}{2} \times 10^{-2} \right)}$$

$$= 16 \times 10^{-5} = 160 \mu T$$

Q6



$$\frac{\mu_0 i}{2R_2} \left(\frac{\pi}{2\pi} \right) \otimes + \frac{\mu_0 i}{2R_1} \left(\frac{\pi}{2\pi} \right) \otimes$$

$$\left(\frac{\mu_0 i}{4R_2} + \frac{\mu_0 i}{4R_1} \right) \otimes$$

$$\frac{4\pi \times 10^{-7} \times 4}{4 \times 4\pi} + \frac{4\pi \times 10^{-7} \times 4}{4 \times 2\pi}$$

$$= 3 \times 10^{-7} = \alpha \times 10^{-7}$$

$$\alpha = 3$$

Q7

$$V = \frac{\mu_0}{4\pi} \frac{M}{r^2}$$

$$\Rightarrow 1.5 \times 10^{-5} = 10^{-7} \times \frac{M}{(20 \times 10^{-2})^2}$$

$$\Rightarrow M = \frac{1.5 \times 10^{-5} \times 20 \times 20 \times 10^{-4}}{10^{-7}}$$

$$M = 1.5 \times 4 = 6$$

Q8

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Solutions

MathonGo

$$R = \frac{mv}{qB} = \frac{p}{qB} = \frac{\sqrt{2m(KE)}}{qB} = \frac{\sqrt{2mqV}}{qB}$$

$$R \propto \sqrt{m}$$

$$m \propto R^2$$

$$\frac{m_1}{m_2} = \left(\frac{R_1}{R_2}\right)^2$$

Q9

$$q = 4\mu C, \vec{v} = 4 \times 10^6 \hat{j} \text{ m/s}$$

$$\vec{B} = 2\hat{k} \text{ T}$$

$$\vec{F} = q(\vec{v} \times \vec{B})$$

$$= 4 \times 10^{-6} (4 \times 10^6 \hat{j} \times 2\hat{k})$$

$$= 4 \times 10^{-6} \times 8 \times 10^6 \hat{i}$$

$$\vec{F} = 32\hat{i} \text{ N}$$

$$x = 32$$

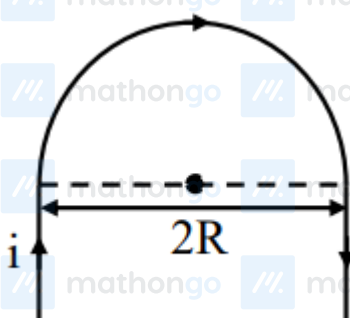
Q10

$$B = 4 \times \frac{\mu_0 i}{4\pi(1/2)} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right)$$

$$= 4 \times 10^{-7} \times 5 \times 2 \times \sqrt{2}$$

$$= 40\sqrt{2} \times 10^{-7} \text{ T}$$

Q11



Note : Direction of magnetic field is in $+\hat{k}$

$$\vec{F} = i\vec{\ell} \times \vec{B}$$

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Solutions

MathonGo

$$\ell = 2R$$

$$\vec{F} = -2iRB\hat{j}$$

Q12

$$\varepsilon = N \left(\frac{\Delta\phi}{t} \right)$$

$$\Delta\phi = (\Delta B)A$$

$$B_i = 5000 \text{ T},$$

$$B_f = 3000 \text{ T}$$

$$d = 0.02 \text{ m}$$

$$r = 0.01 \text{ m}$$

$$\Delta\phi = (\Delta B)A$$

$$= (2000)\pi(0.01)^2 = 0.2\pi$$

$$\varepsilon = N \left(\frac{\Delta\phi}{t} \right) \Rightarrow 22 = N \left(\frac{0.2\pi}{2} \right)$$

$$N = 70$$

Q13

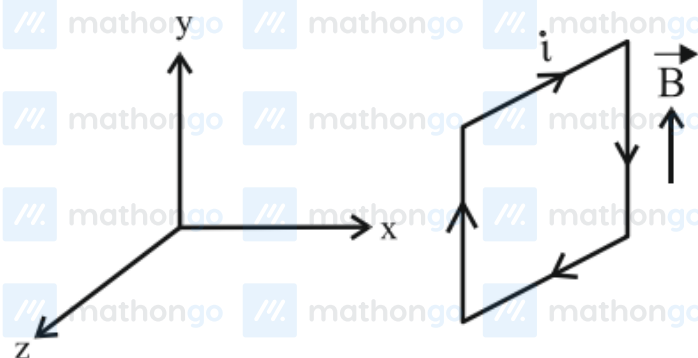
$$\vec{F} = q(\vec{v} \times \vec{B})$$

$$5e\hat{k} = e(3\hat{i} + 5\hat{j}) \times (B_0\hat{i} + 2B_0\hat{j})$$

$$5e\hat{k} = e(6B_0\hat{k} - 5B_0\hat{k})$$

$$\Rightarrow B_0 = 5 \text{ T}$$

Q14



Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

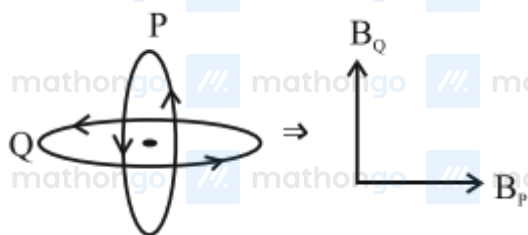
[Click here to download MARKS App](#)

Solutions

MathonGo

$$\begin{aligned}\vec{M} &= i \vec{A} \\ &= 5 \times (0.2) \times (0.1)(-\hat{i}) \\ &= 0.1(-\hat{i}) \\ \vec{\tau} &= \vec{M} \times \vec{B} = 0.1(-\hat{i}) \times (2 \times 10^{-3})(\hat{j}) \\ &= 2 \times 10^{-4}(-\hat{k}) \text{ N-m}\end{aligned}$$

Q15



$$\begin{aligned}B_P &= \frac{\mu_0 N i_1}{2r} = \frac{\mu_0 \times 1 \times 100}{2\pi} = 2 \times 10^{-3} \text{ T} \\ B_Q &= \frac{\mu_0 N i_2}{2r} = \frac{\mu_0 \times 2 \times 100}{2\pi} = 4 \times 10^{-3} \text{ T} \\ B_{\text{net}} &= \sqrt{B_P^2 + B_Q^2} \\ &= \sqrt{20} \text{ mT} \\ x &= 20\end{aligned}$$

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)